

Week 6, Lecture 1: Intro to Maximum Likelihood

Instructor notes. Part 1: prep reading. Part 2: podium cheat sheet

Bring to class:

- **Index cards**, one per student plus a few spares (for the anonymous feedback in Part B).
- A printed copy of the Week 5, Lecture 2 WOD key (in `wod/answer_keys/`).
- Collect last Wednesday's WOD handouts (the handout told students to turn it in today).

This file has two parts, both instructor-only. **Part 1** is read beforehand (worked examples and the reasoning behind the flow). **Part 2** (after the page break, large print) is the podium copy.

There is no WOD today. The period has three beats:

Beat	Time
A. Debrief last Wednesday's WOD	~12 min
B. Course check-in: AI disclosure and anonymous feedback	~20 min
C. Big-picture overview of Bolker Ch. 6	~15 min

Beat B takes the place of the usual end-of-class discussion, so the order differs from a normal day. Putting the feedback in the middle keeps it from being squeezed out if the overview runs long.

Source material: Bolker Ch. 6 intro and 6.2.1 (p. 227-234), plus legacy `LECTURE4.Rmd` ("Likelihood!"), whose MLE section and "why the log-likelihood" discussion are adapted below with a new example.

Where this sits in the week

- **Last week:** Monday built data-generating models (forward: parameters to data). Lab 4 used them for a power analysis, though Act 2 never got started in lab. Wednesday's WOD, done without you in the room, ran the idea backward: nest-fate data held fixed, the likelihood written as a function of p , then logs, a logistic covariate, and a first `dbinom()` negative log-likelihood in R.

- **Today:** close the loop on that WOD, then name the next step. The WOD found the lowest point of the NLL curve by looking at a plot. Maximum likelihood estimation is the formal version of that.
- **Tomorrow (Lab 5):** Act 1 finds an MLE with calculus, on paper. Act 2 fits a model whose MLE has no algebraic solution, using `optim()`.
- **Wednesday:** a live-coding walkthrough of Lab 4 Act 2.

Keep the overview conceptual. Tomorrow’s lab does the derivation, so today should *not* work through the calculus for a binomial MLE (that would give away Act 1). Showing that the calculus route exists is enough.

Part A: WOD debrief (~12 min)

Most of the WOD should need only a quick pass. Spend the time on the four questions that lead into today’s overview. Answers are from the key.

- **2d (do the three likelihoods sum to 1?).** They sum to $46/256 \approx 0.18$, and the area under the whole curve is only $1/20$. Probabilities sum to 1 across possible *outcomes* with p fixed. A likelihood runs across possible *parameter values* with the data fixed, and nothing makes those add to 1. So $\mathcal{L}_4(0.75) \approx 0.105$ is not “the probability that $p = 0.75$.” This is the most important misconception to clear up before tomorrow.
- **3b (1,852 nests).** The product is about 10^{-558} , which underflows to 0 on a computer. The log-likelihood is about -1283.7 , an ordinary number. This is the practical reason for working on the log scale.
- **4c (covariate model vs. constant p).** A likelihood ratio of 2 ($27/128$ vs. $27/256$). Ask what students listed as reasons for caution. The one to highlight: nobody checked whether some other (a, b) pair does better. Finding the best pair is the job of MLE, and that’s the transition to Part C.
- **5c (why the *negative* log-likelihood?).** Take guesses. Answer it properly in Part C, point 2.

Ask briefly what gave people trouble in Part 5 (the R part), since some WOD reflections will say. Save a longer answer for Wednesday’s live coding.

Part B: Course check-in (~20 min)

B1. AI disclosure (~5 min)

These points are a **draft to rewrite in your own words**, not a script. Items in brackets need your confirmation.

- This semester I’m using an AI assistant (Claude) to help redesign the course. It drafts WODs, lab Acts, answer keys, and my own lecture notes.
- The course design is mine: the goals, what goes in each week, the format (WODs, the Acts, pen and paper before laptops). I edit every document, check the math and code,

and rewrite or cut a lot of it.

- Why I'm telling you: one goal of this course is learning to use AI assistants well in an analysis workflow. It would be hypocritical to ask that of you and hide my own use.
- The standard I hold myself to is the one I'm asking of you. The AI drafts, but the person has to understand the result, check it, and take responsibility for it.
- [What stays with me alone, e.g. grading and feedback on your work.]
- If something in a handout looks wrong, say so. The error is mine to fix, whoever drafted it.

Timing note: do the disclosure *before* handing out the cards, so students can respond to it on the card if they want.

B2. Anonymous index cards (~8 min)

Hand out the cards and write the prompts on the board. Say that the cards are anonymous and that honest criticism is more useful to you than politeness, especially about the book, since you're actively considering alternatives.

Side 1, class format. *Start / Stop / Continue:* one thing I should start doing, one I should stop, one I should keep. Anything is fair game: WODs, lectures, lab Acts, pen-and-paper work before laptops, AI use in lab, pacing.

Side 2, the Bolker book.

- In what ways has it been helpful?
- What has been frustrating or confusing about it?
- Roughly how much of the assigned reading do you actually do? (Say plainly that "not much" is a useful answer. The answer shapes whether a different book would even help.)

Collect the cards face down before starting the discussion.

B3. Short open discussion (~7 min)

Cards come before discussion so the first few confident voices don't anchor everyone else's answers. After collecting them:

- **Announce one change already made:** labs now have two shorter Acts instead of three, since we have not finished all three yet.
- Ask one or two open questions, e.g.: *What part of a typical week helps you learn the most? What part feels like busywork?* Or: *When the Bolker reading and the lecture disagree or use different notation, which one do you trust?*
- Don't try to resolve anything today. Say you'll read the cards and report back what you're changing.

Part C: Big-picture overview of Ch. 6 (~15 min)

The WOD introduced likelihood with one example worked in detail. This overview steps back to show the general recipe, using a new example so students see the same moves work beyond the Bernoulli.

Worked example (not from the WOD, the lab, or Bolker): a biologist counts active pika haypiles in six talus plots of equal size and records 2, 5, 3, 0, 4, 4. Model each count as Poisson with an unknown mean λ (haypiles per plot). What value of λ do these data support best?

1. Likelihood, restated (~2 min)

$$\mathcal{L}_6(\lambda) = \prod_{i=1}^6 P(y_i | \lambda) = \prod_{i=1}^6 \frac{\lambda^{y_i} e^{-\lambda}}{y_i!}$$

The data are held fixed and λ varies. $P(y_i | \lambda)$ is a conditional probability of the data. The likelihood itself is written as a function of λ alone, with no bar (the Part A point about 2d).

2. From likelihood to negative log-likelihood (~3 min)

$$\ell_6(\lambda) = \log \mathcal{L}_6(\lambda) = \sum_{i=1}^6 \log P(y_i | \lambda) \quad \text{NLL}(\lambda) = -\ell_6(\lambda)$$

- **Log:** turns the product into a sum, which avoids underflow (WOD 3b) and is easier to differentiate.
- **Negative:** optimization software is written to *minimize*, by convention. Flipping the sign turns “find the peak” into “find the bottom of the valley.” There is no deeper reason than that (answers WOD 5c).

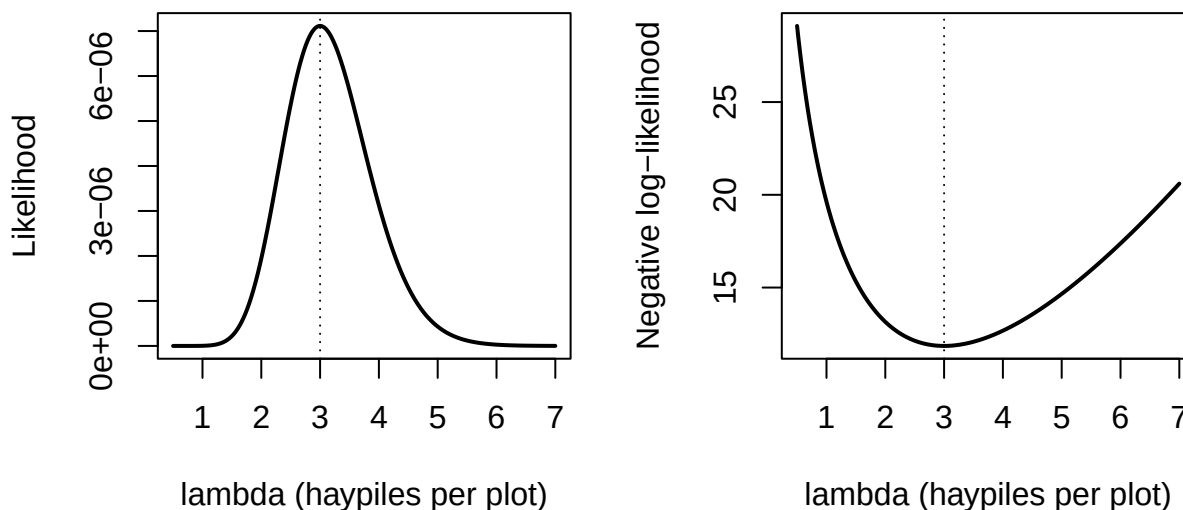
```
haypiles <- c(2, 5, 3, 0, 4, 4)

nll_pois <- function(lambda, y) {
  -sum(dpois(y, lambda = lambda, log = TRUE))
}

lambda_grid <- seq(0.5, 7, by = 0.01)
lik_vals <- sapply(lambda_grid, function(l) prod(dpois(haypiles, l)))
nll_vals <- sapply(lambda_grid, nll_pois, y = haypiles)

par(mfrow = c(1, 2), mar = c(4, 4.5, 1, 1))
plot(lambda_grid, lik_vals, type = "l", lwd = 2,
      xlab = "lambda (haypiles per plot)", ylab = "Likelihood")
abline(v = 3, lty = 3)
plot(lambda_grid, nll_vals, type = "l", lwd = 2,
```

```
xlab = "lambda (haypiles per plot)", ylab = "Negative log-likelihood")
abline(v = 3, lty = 3)
```



3. The maximum likelihood estimate (~2 min)

$$\hat{\lambda} = \arg \max_{\lambda} \mathcal{L}_6(\lambda) = \arg \min_{\lambda} \text{NLL}(\lambda)$$

The two panels above peak and bottom out at the same place, $\lambda = 3$, because the log never changes which of two numbers is bigger (it's monotonic). Here $\hat{\lambda}$ equals the sample mean ($18/6 = 3$). Don't generalize that out loud: it happens to hold for the Poisson, and tomorrow's Act 1 asks students to derive the analogous result for a binomial themselves.

Sticking point to flag: students focus on the *height* of the curve (e.g. “the likelihood at the peak is only about 7×10^{-6} , so this is a bad estimate”). The height depends on how many data points there are and means almost nothing alone. What matters for estimation is *where* the peak is. Whether the curve is sharp or broad matters for uncertainty, which comes later in the semester.

4. Three ways to find the MLE (~6 min)

Bolker's framing: the answer is always the same point, and the methods differ in how they get there.

1. **Calculus.** Differentiate $\ell(\lambda)$, set it to zero, solve. Exact, but only works when the algebra cooperates. (Tomorrow, Act 1.)
2. **Brute-force grid.** Evaluate the NLL at many candidate values and keep the smallest. Easy to understand and to check, but slow, and hopeless with many parameters (a 100-point grid in each of 5 parameters is $100^5 = 10^{10}$ evaluations). This is what the WOD's `supply()` plot did.

3. **Numerical optimizer.** An algorithm starts from a guess and walks downhill. R's `optim()` needs the NLL function and a starting value. (Tomorrow, Act 2.)

```
lambda_grid[which.min(nll_vals)]           # grid

## [1] 3

fit <- optim(par = 2, fn = nll_pois, y = haypiles,
             method = "Brent", lower = 0.01, upper = 20)
fit$par                                     # optimizer

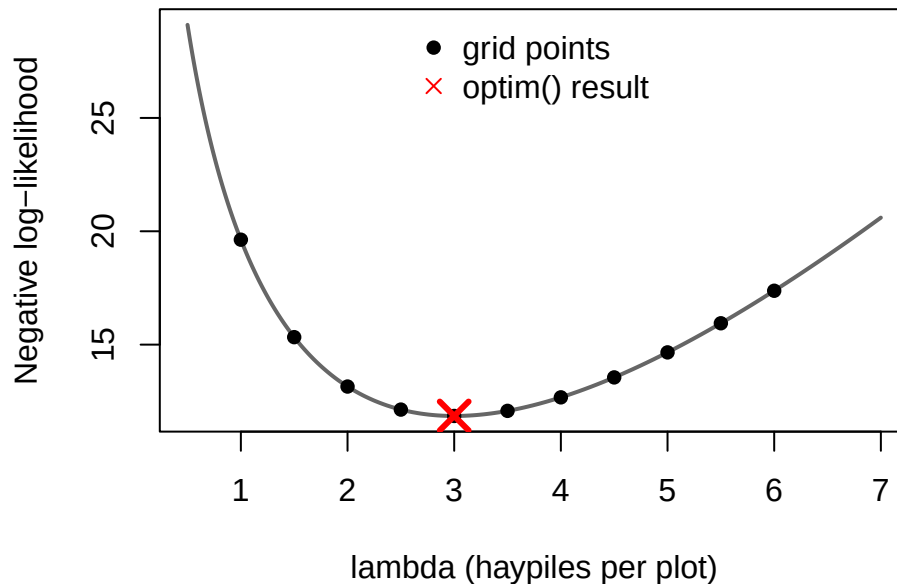
## [1] 3

fit$value                                  # NLL at the MLE

## [1] 11.85348
```

Both land at 3 (up to rounding). `method = "Brent"` is `optim()`'s option for a single parameter. With two or more parameters the default method works, which is the setting for Act 2.

```
coarse <- seq(1, 6, by = 0.5)
par(mar = c(4, 4.5, 1, 1))
plot(lambda_grid, nll_vals, type = "l", lwd = 2, col = "grey40",
     xlab = "lambda (haypiles per plot)", ylab = "Negative log-likelihood")
points(coarse, sapply(coarse, nll_pois, y = haypiles), pch = 16)
points(fit$par, fit$value, pch = 4, cex = 2, lwd = 3, col = "red")
legend("top", legend = c("grid points", "optim() result"),
     pch = c(16, 4), col = c("black", "red"), bty = "n")
```



5. Teaser: continuous data (~2 min)

For a continuous response the likelihood uses the probability *density* instead of a probability, e.g. `dnorm()` in place of `dpois()`. Densities can exceed 1, so a likelihood can too:

```
dnorm(0, mean = 0, sd = 0.2)    # a density above 1
```

```
## [1] 1.994711
```

This is fine. Only the location of the peak (and comparisons between likelihoods on the same data) carry meaning, so the scale was never interpretable anyway. Bolker 6.2.1 (p. 234-236) fits a Gamma model to myxomatosis virus titers this way.

Take-home message

- A likelihood is the probability of the observed data, viewed as a function of the parameters. It is not a probability distribution over the parameters.
- We minimize the negative log-likelihood: the log for numerical and algebraic convenience, the minus sign because optimizers minimize.
- The MLE is the parameter value at the bottom of the NLL curve. Calculus, grids, and optimizers are three routes to the same point.

PART 2: PODIUM CHEAT SHEET

Terse on purpose. Full rationale and code: Part 1.

Bring: index cards. **Collect:** last Wednesday's WOD.

A. WOD debrief (~12 min)

- **2d:** sums to 0.18, not 1. Likelihood runs over *parameters*, data fixed. $\mathcal{L}_4(0.75)$ is not “ $P(p = 0.75)$ ”
- **3b:** product $\approx 10^{-558}$ underflows to 0; log sum ≈ -1284
- **4c:** ratio of 2, but nobody searched for the *best* (a, b) . That search is MLE
- **5c:** why negative? Hold the answer for Part C
- Quick poll: what went wrong in the R part?

B. Course check-in (~20 min)

B1. AI disclosure (~5 min). Own words. Points:

- Claude drafts WODs, Acts, keys, my notes
- Design is mine; I edit and check everything
- Course goal includes using AI well, so it would be hypocritical to hide it
- Same standard as for you: understand it, check it, own it
- Found an error? Tell me

B2. Index cards (~8 min). Anonymous. Honest beats polite.

- Side 1: **Start / Stop / Continue** (WODs, lectures, Acts, laptops, AI in lab, pacing)
- Side 2, **Bolker**: helpful? frustrating or confusing? how much reading do you actually do?
- Collect face down

B3. Discussion (~7 min).

- Announce: labs now 2 shorter Acts
- What helps you learn most? What feels like busywork?
- Don't resolve; promise to report back

C. Overview of Ch. 6 (~15 min)

Pika haypiles in 6 plots: **2, 5, 3, 0, 4, 4**. $\text{Poisson}(\lambda)$.

1. Likelihood (~2 min)

$$\mathcal{L}_6(\lambda) = \prod_{i=1}^6 \frac{\lambda^{y_i} e^{-\lambda}}{y_i!}$$

- Data fixed, λ varies. No bar on $\mathcal{L}$

2. NLL (~3 min)

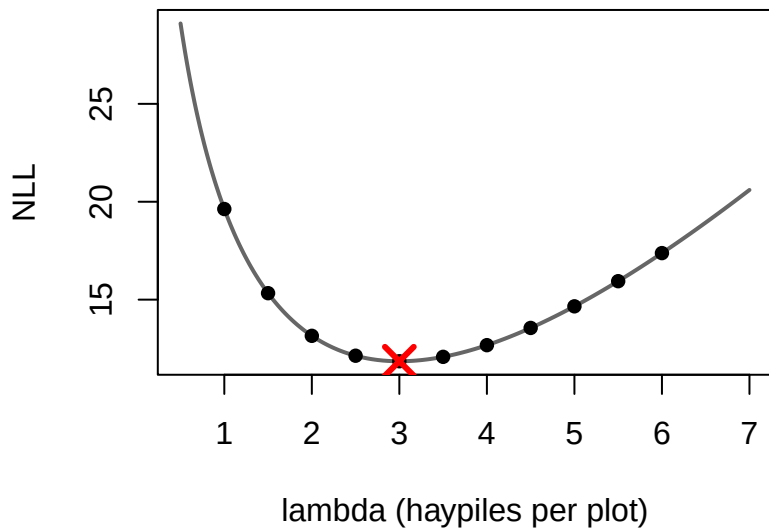
- Log: product to sum, no underflow, easier calculus
- Minus: optimizers minimize (answers WOD 5c)

3. MLE (~2 min)

$$\hat{\lambda} = \arg \min_{\lambda} \text{NLL}(\lambda) = 3$$

- Same location on $\mathcal{L}$, ℓ , NLL (log is monotonic)
- **Sticking point:** location matters, not height

4. Three routes to the MLE (~6 min)



- **Calculus:** $d\ell/d\lambda = 0$. Exact when algebra allows (Lab Act 1)
- **Grid:** easy, slow; $100^5 = 10^{10}$ for 5 parameters (the WOD plot)
- **Optimizer:** start from a guess, walk downhill; `optim()` (Lab Act 2)

5. Teaser (~2 min)

- Continuous data use densities: `dnorm(0, 0, 0.2)` ≈ 2
- Likelihood can exceed 1; only peak location and comparisons matter

Tomorrow: Lab 5. Calculus, then `optim()`.